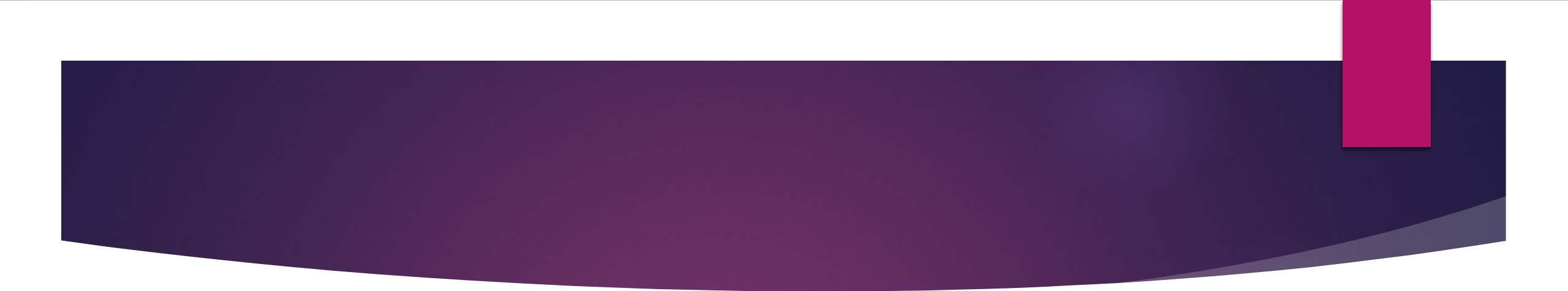


Information System in Global Business Today

-NILAM SANGROULA

Information Systems: Definition

- ▶ Information system has been defined in terms of two perspectives: one relating to its function; the other relating to its structure.
 - ▶ **Functional perspective** of information system is a technologically implemented medium for the purpose of recording, storing, and disseminating linguistic expressions as well as for the supporting of inference making.
 - ▶ **Structural perspective** of information system consists of a collection of people, processes, data, models, technology and partly formalized language, forming a cohesive structure which serves some organizational purpose or function.
- ▶ The *functional* definition has its merits in focusing on what actual users do with the information system while using it. They communicate with experts to solve a particular problem. The *structural* definition makes clear that IS are socio-technical systems, i.e., systems consisting of humans, behavior rules, and conceptual and technical artifacts.

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- ▶ An information system is a collection of many sets of data that ensures the successful completion of a business objective. Information systems are not a stand-alone model of the IT industry. Instead, a crucial aspect of their successful implementation is their integration with data and business processes.
 - ▶ Thus, it is better to think of the information system as a triangle. The three parts of this triangle represent processes, people and computers. An information system requires each of these components to work properly to succeed.

Tools used in an information system

- ▶ An information system uses a range of equipment to allow for the effective distribution of information. This equipment often includes tools, such as different forms of hardware, or virtual equipment composed of different systems and application software.
- ▶ An easy way to understand the information system is to think of it as a software program that helps to keep data organized and perform various types of analysis on the data. Usually, it facilitates the conversion of raw data into useful information.
- ▶ For example, consider a survey that has collected data about multiple products bought by teenagers. An information system not only helps to sort this data, but it also helps to perform data analysis. Such analysis may reveal trends of buying habits in teenagers or the products that are currently in popular demand.

Types of information systems

- ▶ The type of system the organization uses will depend on its current and overall business objectives.
- ▶ **Operations support systems**
 - ▶ Operations support systems complement a certain operation of the business. An example of an operations support system could be the transaction processing systems used in almost all banks around the world. Usually, this type of software allows the service provider to analyze a particular process of the business. They also help in controlling and managing a service in a specific network.
- ▶ **Management information systems**
 - ▶ The second major category of information systems is management information systems. These systems usually consist of a combination of hardware and software integration that help an organization perform its core functions. Thus, they are involved in gathering data from multiple online systems. This information is not only stored by such systems but is also analyzed and reported in a useful manner to help management make decisions for their organization.

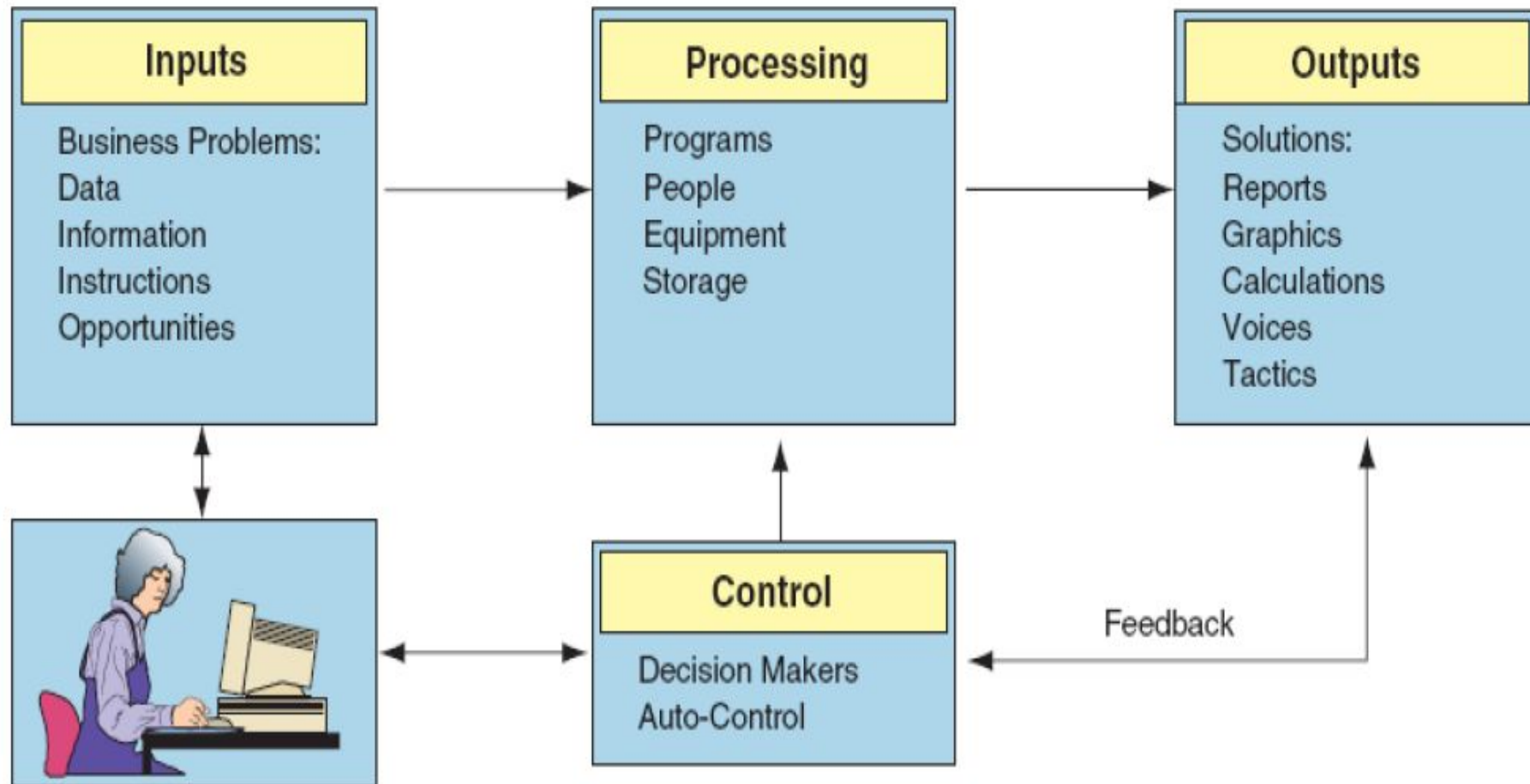
▶ **Decision support systems**

- ▶ The decision support systems help an organization to make informed decisions regarding its operations. Often, the focus of these systems is to analyze information that is rapidly changing and cannot be determined easily in advance. Such systems are available both in the form of completely automated systems or completely human-operated systems. However, the trend in DSS has been a combination of human and computer-operated systems for maximum efficiency.

▶ **Executive information systems**

- ▶ The last major category of information systems is the executive information systems, commonly abbreviated as ESS. These information systems serve as management support systems. Usually, they are used in the senior-level decision-making needs of an organization. They are usually also considered an advanced form of DSS.

Functions of an information system



Management Information System

- ▶ Management information systems (MIS) are an organized method of collecting information from various sources, compiling it, and presenting it in a readable format. It helps business leaders and managers make strategic management decisions.
- ▶ Management Information System (MIS) focuses on managing and using specific information systems to support operational, tactical, and strategic decision-making within an organization. It primarily deals with collecting, storing, processing, and disseminating information to facilitate managerial activities.

Types of MIS

Type of MIS	Description
Process Control	Gather data to create reports based on the performance of systems and processes.
Management Reporting System	Generate reports for the company's operations.
Inventory Control	Allow tracking of the current inventory state within a department or the company.
Decision Support Systems	Gather information from internal and external resources and help team management make efficient business decisions.

Type of MIS	Description
Executive Information System	Report company data to top management directly in an easy-to-read format.
Transaction Systems	Automate business processes and collect data on a company's daily transactional activities.
Accounting & Finance Systems	Track a company's assets and investments and processes financial and accounting-related operations.
Sales & Marketing Systems	Facilitate tracking of a company's sales and marketing efficiency.

Functions of MIS

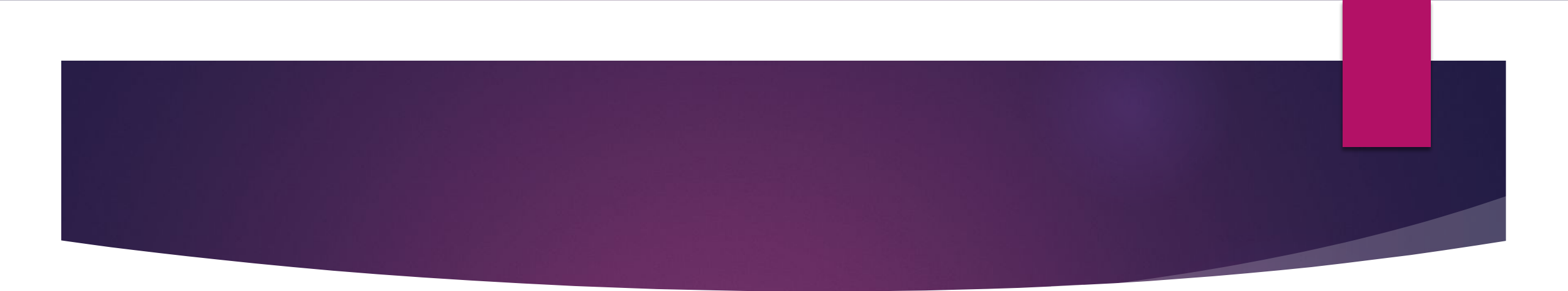
- ▶ The primary function of MIS is to report on business operations to support decision-making and ensure that the organization is managed more efficiently. This will help the company reach its full potential and thus gain a competitive advantage. Listed below are some of the crucial functions of MIS.
 - ▶ Provide Easy Access to the Information
 - ▶ Data Collection
 - ▶ Performance Tracking
 - ▶ Foster Collaboration in the Workplace
 - ▶ Company Projections
 - ▶ Track the Implementation of Decisions
 - ▶ Improve Company Reporting

Limitations of MIS

- ▶ Even though MIS has many benefits, it also has its limitations, which are discussed below:
- ▶ While MIS may solve some acute problems, it is not a solution to all problems of an organization.
- ▶ Involves maintenance and employee training costs.
- ▶ It cannot meet the particular demands of everyone.
- ▶ If incorrectly designed, MIS does not serve the management and is irrelevant.
- ▶ The MIS is only good if the primary data is updated.
- ▶ Most information provided by the MIS is in quantitative form. Hence, it ignores qualitative information like the behavior of an employee.

Globalization challenges and opportunities of IS

- ▶ Globalization refers to an open flow of information, technology, and goods among countries and consumers. This openness occurs through various relationships, from business, geopolitics, and technology to travel, culture, and media.
- ▶ Challenges of Globalization
 - ▶ Diverse Regulatory Compliance
 - ▶ Cultural and Language Barriers
 - ▶ Cybersecurity Risks
 - ▶ Data Privacy Concerns
 - ▶ Supply Chain Disruptions
 - ▶ Talent Management and Skill Gap

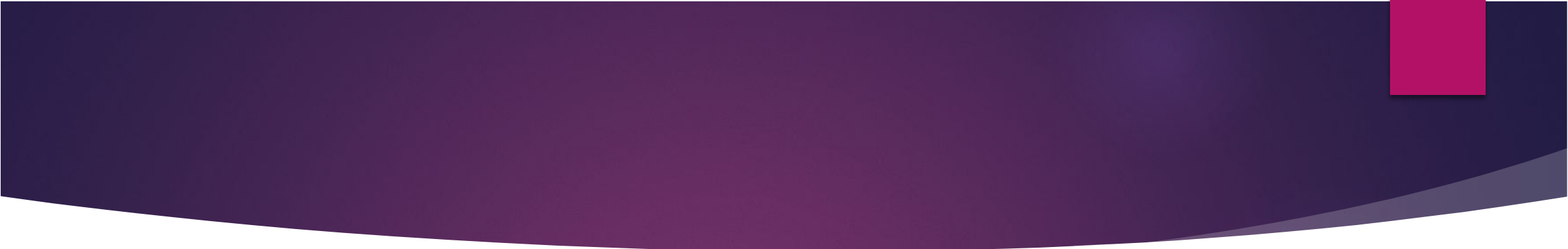
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- ▶ Opportunities of Globalization
 - ▶ Access to a Larger Market
 - ▶ Diverse Talent Pool
 - ▶ Cost Optimization
 - ▶ Collaboration and Knowledge Sharing
 - ▶ Technology Innovation and Learning
 - ▶ Strategic Alliances and Partnerships

Strategic Business objectives of IS

- ▶ Information systems are a necessity of doing business in the 21st century. Moreover, information systems enable organizations to thrive and survive during turbulent times. As a result, organizations invest in information systems to achieve six strategic business objectives:
 - ▶ Achieve operational excellence
 - ▶ Create new products, services, and business models
 - ▶ Improve customer and supplier intimacy
 - ▶ Improve decision-making
 - ▶ Establish competitive advantage; and
 - ▶ Survive.

Contemporary Approach to IS

- ▶ The following examples show how the study of information systems is a multi-disciplinary area that draws on theories from several disciplines: The study of IS focuses on problems that have internal contributions from the technical and behavioral sciences.
- ▶ **Technical Approach**
 - ▶ The mathematically based model is prioritized in the technical approach to information systems.
 - ▶ The actions are carried out in order.
 - ▶ Only after the prior activity is finished is the subsequent activity started.
 - ▶ Computer science, management science, robotic science, and operational research science are the disciplines that support the technical approach.
 - ▶ The goals of computer science are to establish computability and develop effective data storage techniques.
 - ▶ The development of models for decision-making and management practices is emphasized in management science.
 - ▶ Operation research focuses on mathematical methods for enhancing specific organizational parameters like transportation, inventory management, and transaction costs.

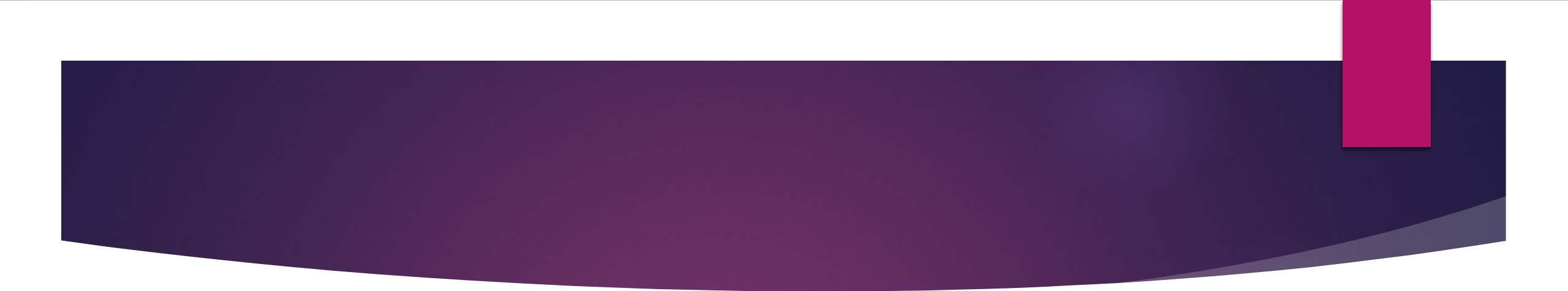


▶ **Behavioral Approach**

- ▶ Various behavioral challenges, including strategic business integration, design, execution, and utilization, are included in the information system sector.
- ▶ Although the behavioral approach does not completely disregard technology, it often places more of an emphasis on technical solutions.
- ▶ Sociology, psychology, and economics are the three disciplines that make up a behavioral approach.
- ▶ Sociologists focus their research on information systems on how groups and organizations create systems and how systems affect particular groups and organizations.
- ▶ When evaluating information systems, psychologists consider how formal information is viewed and used by human decision-makers.
- ▶ Economists are interested in understanding how digital goods are produced, the dynamics of the digital market, and how new information systems alter and regulate the cost structure within a corporation.

Business Perspectives of IS

- ▶ Information Systems (IS) are integral to modern businesses, offering a multitude of perspectives that influence various aspects of operations, management, and strategic decision-making. Here are key business perspectives of IS:
 - ▶ Operational Excellence
 - ▶ Customer Service and Experience
 - ▶ Strategic Advantage and Innovation
 - ▶ Business Model Transformation
 - ▶ Data-Driven Decision Making

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- ▶ Risk Management and Compliance
 - ▶ Supply Chain Optimization
 - ▶ Knowledge Management and Learning
 - ▶ Financial Management and Reporting
 - ▶ Globalization and International Expansion
 - ▶ Employee Productivity and Satisfaction